

Conclusion

The results of this investigation demonstrate that the original hypothesis was not supported. It was predicted that WIWT systems near the trailing edge, would generate supplemental electrical power with minimal aerodynamic penalty. However, this assumption proved incorrect because the drag created by turbine integration consistently required more propulsion power to overcome than the turbine could generate in return.

Beyond the rejection of the hypothesis, several key findings emerged. First, geometric thickness proved to be the strongest predictor of power output, acting as a velocity amplifier that increased local airflow speed and therefore turbine performance. Second, camber did not provide meaningful aerodynamic protection against integration penalties; lift degradation occurred consistently regardless of curvature. Third, internal placements restricted mass flow and reduced turbine effectiveness due to stagnation effects. Fourth, turbine placement along the chord followed a distinct aerodynamic pattern in which mid-chord integration minimized lift loss and mechanical volatility compared to forward or rearward extremes. Finally, mechanical feasibility depended not only on RPM but on maintaining a stable torque-to-speed ratio to avoid structural fatigue. However, across every configuration, the aerodynamic penalty from induced drag exceeded the electrical power generated by the turbines, resulting in a net energy deficit. Together, these findings confirm that while WIWT systems are mechanically capable of efficient energy extraction, aerodynamic compatibility remains the central obstacle to profitable cruise integration.

To address this challenge, one realistic solution is the strategic use of regenerative energy capture during descent and landing. Aircraft are required to generate drag in these phases to reduce speed and altitude. Integrating WIWT systems into this operational window would

transform unavoidable drag into productive energy recovery. Instead of functioning as a propulsion burden, the turbines would operate when aerodynamic resistance is already necessary. The recovered electricity could power onboard systems or recharge batteries for hybrid aircraft. This solution benefits overall system efficiency because it aligns turbine operation with natural flight dynamics rather than opposing them. A limitation, however, is that descent represents only a fraction of total flight time. The energy captured may not fully offset the added structural weight carried throughout climb and cruise, making lightweight materials and efficient storage essential for net benefit.

A second solution involves retractable WIWT architectures combined with adaptive flow management. By allowing turbines to deploy only during surplus-energy states and remain sealed during cruise, the aircraft can avoid continuous aerodynamic disruption. Supplementing this system with active flow control would smooth airflow around the aperture when deployed, reducing separation and drag spikes. This approach enhances overall viability because it preserves aerodynamic efficiency during steady flight while still enabling energy harvesting when conditions permit. The limitation lies in structural and mechanical complexity. Moving components, reinforcement structures, and control systems introduce new failure points and maintenance demands, potentially increasing certification difficulty and long-term operational cost.

Ultimately, this research shows that the barrier to effective airborne energy harvesting is not the turbine's ability to spin efficiently, but the wing's ability to remain aerodynamically optimized while carrying it. WIWT technology demonstrates strong mechanical promise, yet successful implementation requires designs that cooperate with airflow rather than disrupt it. Future research should focus on variable-geometry airframes, lightweight composite integration

strategies, and broader flight-envelope simulations to determine where energy recovery can complement, rather than compete with, aerodynamic performance. By identifying the NACA 23015 T-M configuration as a baseline, this study provides the mathematical foundation for a future where aircraft do not just consume energy, but actively recover it from the very air they traverse.